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ENGINEERING



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Evolution of Natural Language Processing

From Rule-Based Systems to Modern NLP

DSA0305-Natural Language Processing

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What is Natural Language Processing?

Definition: Natural Language Processing (NLP) enables computers to process and understand human languages such as English, Telugu, Tamil, Hindi, and many others.

1 Objectives

- Understand human language
- Process text and speech
- Extract useful information
- Generate meaningful responses
- Support intelligent decision-making

2 Applications

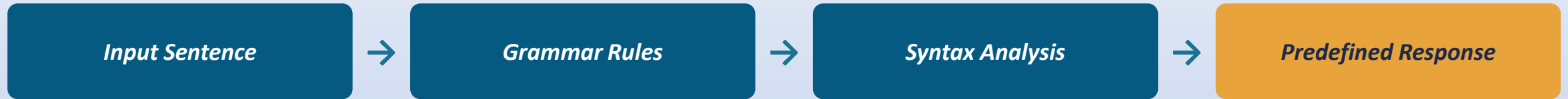
- Chatbots & Virtual Assistants
- Machine Translation
- Speech Recognition
- Sentiment Analysis
- Spam Detection
- Text Summarization
- Grammar Checking
- Search Engines

3 Challenges

- Ambiguity in language
- Context understanding
- Multiple languages
- Slang and abbreviations
- Different writing styles

Rule-Based NLP Systems

The earliest NLP systems relied on manually written grammar rules and dictionaries created by language experts.



1 Characteristics

- Handwritten grammar rules
- Pattern matching
- Dictionary lookup
- If-Else logic
- Domain-specific knowledge

2 Advantages

- Easy to understand
- Predictable outputs
- High accuracy in limited domains
- No training data required

3 Limitations

- Difficult to maintain
- Cannot learn from data
- Poor scalability
- Unable to understand unseen sentences

Statistical NLP

Statistical NLP replaced manual rules with probability-based methods that learn language patterns from large text datasets.

1 Techniques

- N-Gram Models
- Hidden Markov Models (HMM)
- Bayesian Networks
- Maximum Entropy Models

2 Advantages

- Learns from data
- Handles language variations
- Better machine translation
- More flexible than rule-based systems

3 Limitations

- Requires large datasets
- Limited context understanding
- Data sparsity problems

4 Applications

- Speech Recognition
- Spell Checking
- POS Tagging
- Machine Translation

Machine Learning-Based NLP

Machine Learning allowed NLP systems to automatically identify language patterns from data instead of depending only on manually written rules.

1 Algorithms

- Decision Trees
- Naïve Bayes
- Support Vector Machines (SVM)
- Random Forest

2 Feature Engineering

- Bag of Words
- TF-IDF
- Word Frequency
- POS Features

3 Advantages

- Better prediction accuracy
- Faster model development
- Improved text classification
- More adaptable to new datasets

4 Limitations

- Manual feature engineering required
- Weak context understanding
- Performance depends on feature quality

Deep Learning & Modern NLP

Deep Learning transformed NLP by automatically learning language features and understanding context using neural networks.

1 Neural Networks

- RNN
- LSTM
- GRU
- Transformer Architecture

2 Transformer Revolution

The Transformer model introduced in 2017 enabled faster training and better context understanding using the attention mechanism.

3 Popular Models

- BERT
- GPT
- RoBERTa
- T5
- XLNet

4 Advantages

- Understands long context
- Generates human-like text
- High accuracy
- Supports multiple NLP tasks

CONCLUSION

- 1 NLP has evolved from rule-based systems to advanced deep learning and Transformer-based models, significantly improving language understanding.
- 2 Statistical and Machine Learning techniques enabled computers to learn from data, reducing dependence on manually written rules.
- 3 Modern NLP models such as BERT and GPT can understand context, generate human-like text, and perform multiple language tasks with high accuracy.
- 4 NLP is widely used in chatbots, virtual assistants, translation systems, search engines, healthcare, finance, and education, making technology more accessible and efficient.
- 5 Future advancements in NLP will focus on multilingual support, ethical AI, explainability, and more intelligent human-computer interactions, driving innovation across various industries.

THANK YOU!!